Engineering - Graduate Level Assessment 10 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Find the radiation resistance of a dipole antenna (1 / 10) wavelengthlong.
   (a) 7.9 ohms
   (b) 3.9 ohms
   (c) 10.2 ohms
   (d) 6.3 ohms
   (e) 12.5 ohms

2. If a body is cooled below ambient temperature by a system, it can then be used to freeze water. Calculate the maximum air temperature for which freezing takes place.
   (a) 190
   (a) 200
   (a) 170
   (a) 180
   (a) 120

3. Find the current through a 0.01 capacitor at t = 0 if the voltage across it is (a) 2 sin 2π 10^6 t V; (b) - 2 $e^-(10)7tV; (c) - 2e - (10)7tsin2\pi10^6$ t V.
   (a) 0.1256 A, 0.2 A, 0.1256 A
   (b) 0.1256 A, 0.3 A, 0.1256 A
   (c) 0.2345 A, 0.1 A, - 0.4567 A
   (d) 0.2 A, 0.1 A, 0.2 A
   (e) 0 A, 0.1 A, 0 A

4. Find the general solution of Bessel's equation of order one.
   (a) y(x) = c_1 * sin(x) + c_2 * cos(x)
   (b) y(x) = c_1 * y_1$(x)^2$ + c_2 * y_2$(x)^2$
   (c) y(x) = c_1 y_1(x) + c_2 y_2(x)
   (d) y(x) = c_1 * x * y_1(x) + c_2 / (x * y_2(x))
   (e) y(x) = c_1 * y_1(x) / (c_2 * y_2(x))

5. The errors mainly caused by human mistakes are

(a) gross error.
(b) instrumental error.
(c) systematic error.

True / False (10 marks)

Answer True or False

6. True or False: A correction of -10 mph must be applied to the constant-density indication when an airplane is cruising at 300 mph.

7. True or False: The sequence y[n] satisfying the given recurrence relation is y[n] = 6 - $(1/2)^n$ - $(1/3)^n$.

8. True or False: The regulation of a 2,300 volt 60-cycle power transformer using the per-unit method is 2.25 percent.

9. True or False: The delta response h[n] of the system is h[n] = 3 $(1/2)^n$ u[n] - 2 $(1/2)^n$ u[n].

10. True or False: The probability of rolling a 'six' with a fair six-sided die is 1/6.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the power delivered by a roller chain with a 3/4 in. pitch on a sprocket with 34 teeth rotating at 500 rpm, and discuss the implications of your findings in mechanical systems.

12. Given a liquid in a cylinder with a volume of 1 liter at 1 MN/m^2 and 995 cm^3 at 2 MN/m^2, calculate the bulk modulus of elasticity and discuss its significance in engineering applications.

End of Paper